Yogesh Sharma

Roll No.: B22CH045
B. Tech in Chemical Engineering
Minor in Artificial Intelligence & Data Engineering
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Education

Degree/Certificate	Institute	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology Jodhpur	7.79	2026
CBSE (XII)	Maheshwari Public School, JN, Jaipur	92.0%	2022
CBSE (X)	Maheshwari Public School, JN, Jaipur	96.0%	2020

Experience

- **Thuriyam AI, Bangalore** *Dec'25 - Mar'26*
  AI & Backend Engineer Intern | Skills- Natural Language Processing, Multimodal LLMs Remote
  - Engineered an automated transcription pipeline using **FastAPI** and **Gemini models** deployed on **cloud infrastructure (AWS)** to process **10K+ daily call recordings** across sales and support teams at **WorkIndia**.
  - Architected end-to-end NLP analysis pipelines to extract actionable business intelligence including **competitor mentions**, **top-feature requests**, and **product challenges** currently utilized across **product and sales operations**.
  - Reduced LLM API costs by **97%** on **competitor analysis** workflows by implementing a deterministic **subword pre-filtering engine**, routing only the **3%** of relevant transcripts to heavy AI analysis without quality loss.
  - Sustained **>99.98% reliability** across **300,000+ monthly requests**, diagnosing production edge cases and reducing operational errors to under **50 per month**; deployed **Dockerized microservices** with **Jenkins CI/CD** automation.
- **NLP Lab, Indian Institute of Science Bangalore (IISc)** *May'25 - Aug'25*
  NLP Research Intern | Skills- Deep Learning, Natural Language Processing, Embeddings, LLMs Hybrid
  - Investigated methods to enhance **LLM-based encoders** for **semantic similarity and retrieval tasks** with reduced computational cost, with potential implications in downstream **recommendation systems**
  - Created a **180K+ sample cross-lingual** embedding dataset using **NV-Retriever–based hard negative mining**, generating **10 hard negatives** per anchor with **e5-Mistral-7B-Instruct** for contrastive learning.
  - Achieved a **5%** improvement on **Semantic Textual Similarity (STS)** benchmarks and **2%** improvement on **Retrieval** benchmarks while training only **0.001%** of **model parameters** using soft-prompt tuning.
- **AI Stealth Startup, Bangalore** *Aug'25 - Nov'25*
  Data Scientist Intern | Skills- Text mining, Big data, Generative AI, Conversational agents Remote
  - Curated a **150K-sample multi-label safety dataset** (toxic, NSFW, partial-toxic, code, self-harm, illegal, prompt-injection, safe), with **30K samples** reserved for testing using data-mining techniques to train lightweight models.
  - Trained a **BERT-based guardrail model**, achieving **95.8% accuracy / 0.96 micro-F1 / 0.74 macro-F1, with <50 ms inference latency**, **95%+ faster** than LLM-based guardrails.
  - Evaluated per-class safety performance, maintaining **>0.83 F1 on NSFW**, **0.63 on prompt-injection**, and **0.99+ on self-harm** and **illegal content**, ensuring robust real-time safety filtering.

Projects

- **Inter IIT Tech Meet 13.0** *Oct'24 - Dec'24*
  Skills- Data science, Predictive modeling, Statistical modeling, data mining | Report Link
  - Engineered scalable **ETL** and feature-engineering pipelines using **Pandas** and **PySpark**, transforming ball-by-ball match data into match-level features and integrating role metadata for **2,500 players**.
  - Built **3 specialized analytical datasets** capturing **player form**, **venue effects**, and **player-vs-player** matchup dynamics using up to **10 years** of historical context for predictive modeling.
  - Benchmarked **LightGBM**, **XGBoost**, **Random Forest**, **SVR**, **Polynomial** and **Linear Regression** across T20, ODI and Test formats; selected explainable tree-based models for robust non-linear predictive modeling, achieving **27 MAPE** and ranking **6th among 23 IITs** (top 26%).
  - Applied hypothesis testing, feature-importance analysis and **SHAP-based explainability** to identify performance drivers and translate player-level predictions into interpretable insights; integrated Qwen-based GenAI explanations.

Technical Skills

–**Programming**: Python, C/C++
–**Frameworks / Libraries**: Numpy, Pandas, Matplotlib, OpenCV, scikit-learn, PyTorch, Hugging Face, Tensorflow
–**Cloud & Infrastructure**: PySpark, Kafka, AWS, SQL, Docker, Jenkins, Hadoop
–**Tools & OS**: Git/GitHub, Linux, Windows, Jupyter Notebooks

Relevant Coursework

– Andrew Ng's ML and DL Specialization | Introduction to ML | Deep Learning | Data Structures & Algorithms
– Probability Statistics and Stochastic Processes | Mathematics I (Calculus) | Mathematics II (Linear Algebra

Achievements

– **6th Position out of 23 IITs**, Dream11 Problem Statement, Inter-IIT Tech Meet 13.0, IIT Bombay *2024*
– **Among top 10**,ML Kaggle Challenge worldwide, by RAID. *2023*